# Computer Lab 1, Assignment 2

Linear regression and ridge regression

## Assignment 2: Linear regression and ridge regression

### 1. Data partitioning and scaling

The data were split 60/40 into training and test sets with the code from Lecture 1a. They were then scaled with `preProcess`, using means and standard deviations from the **training data only**, applied to both sets. Only the 16 voice characteristics are used as features and `motor_UPDRS` as target. No intercept is used, since the data are centred. MSE values are in standardized units.

### 2. Linear regression

```
fit_lm <- lm(motor_UPDRS ~ 0 + ., data = trainS)
lm_train_mse <- mean((y - predict(fit_lm, trainS))^2)
lm_test_mse  <- mean((ytest - predict(fit_lm, testS))^2)
```

**Training MSE = 0.8785, test MSE = 0.9354.**

Variables significant at the 5% level (smallest p-value first): DFA, PPE, HNR, Shimmer.APQ11, Jitter.Abs., NHR, Shimmer.APQ5, Shimmer. Several voice measures are near copies of each other (e.g. `Jitter.DDP` = 3 x `Jitter.RAP`), so collinearity inflates standard errors and a non-significant variable is not necessarily unimportant. Both MSEs are fairly high, so the voice measures explain only part of the variation in `motor_UPDRS`.

### 4. Ridge regression for $\lambda = 1, 100, 1000$

Table 1: MSE and degrees of freedom (lambda = 0 is the linear regression from step 2).

| lambda | train_MSE | test_MSE | DF |
|---|---|---|---|
| 0 | 0.8785 | 0.9354 | 16.0000 |
| 1 | 0.8786 | 0.9350 | 13.8607 |
| 100 | 0.8844 | 0.9323 | 9.9249 |
| 1000 | 0.9211 | 0.9539 | 5.6439 |

**Most appropriate penalty.** Judged by test MSE, the best of the three is $\lambda = \mathbf{100}$ (test MSE 0.9323, versus 0.9354 without penalty). The training MSE increases with $\lambda$ and cannot be used for this choice.

**Degrees of freedom.** DF is 13.86, 9.92, 5.64 for $\lambda = 1, 100, 1000$, compared with 16 without penalty. DF falls as $\lambda$ grows: the coefficients are shrunk, so the model becomes simpler. A moderate penalty helps with the collinear features, while too large a penalty underfits.